

Team AT_{STA_mono:72} (Wei, 2024) adopts three different semantic embedding algorithms, GLOVE, n-gram embeddings and SentenceBERT as well as their concatenation. The author finds that these pre-trained embeddings, while fast to compute, are not as effective as a fine-tuned RoBERTa model.

Team DUTh_{STA_mono:73} (Kyriakou et al., 2024) experiments with several supervised classification models based on PLMs. Finally, they opt for a fine-tuned mBERT trained for 5 epochs. This approach shows how PLMs fine-tuning is a versatile approach that can be effective when detecting synthetic texts.

Team surbhi_{STA_mono:74} (Sharma and Mansuri, 2024) creates two sets of features (a) stylometric features based on the length of text, the number of words, the average length of words, the number of short words, the proportion of digits and capital letters, individual letters and digits frequencies, hapax-legomena, a measure of text richness, and the frequency of 12 punctuation marks and (b) n-grams: frequencies of the 100 most frequent character-level bi-grams and tri-grams; (c) the output probabilities of fine-tuned Roberta model. Each set of features is used to train a classifier and finally, stylometric and n-gram features are chosen as the best-performing ones. They prove that more classical features can still be valuable when attempting the detection of synthetic text.

Team Kathlalu_{STA_mono:76} (Cao et al., 2024) investigates two methods for constructing a binary classifier to distinguish between human-generated and machine-generated text. The main emphasis is on a straightforward approach based on Zipf's law, which, despite its simplicity, achieves a moderate level of performance. Additionally, they briefly discuss experimentation with the utilization of unigram word counts.

Team KInIT_{STA_mono:77} (Spiegel and Macko, 2024) uses two approaches: (a) an ensemble using two-step majority voting for predictions, consisting of 2 LLMs (Falcon-7B and Mistral-7B) fine-tuned using the train set only; (b) 3 zero-shot statistical methods (Entropy, Rank, Binoculars) using Falcon-7B and Falcon-7B-Instruct for calculating the metrics. For classification they use per-language threshold calibration, showing that likelihood-based methods are a viable solution to detect machine-written texts.

Team iimasNLP_{STA_mono:78} (Valdez et al., 2024) fine-tune 4 different language models to identify human and machine generated text, ERNIE, SpanBERT, ConvBERT and XLNet. They find out that RoBERTa is a stronger classifier. In general this shows how fine-tuning PLMs is an effective approach to identify synthetic text.

Team Groningen-F_{STA_mono:86} (Donker et al., 2024) leverage features including tense of the sentence, the voice of the sentence, the sentiment of the sentence, and the number of pronouns vs. proper nouns on the basis of SVM and FFNN models. The hypothesis here is that traditional models may generalize better than LLMs. It is more computationally effective than LLMs.

Team RUG-D_{STA_mono:100} (Brekhof et al., 2024) fine-tunes different DeBERTa models on a dataset extended with additional synthetic samples. Showing that PLMs fine-tuning is a versatile approach that can be effective in the detection of synthetic texts.

Team RUG-5_{STA_mono:101} (Darwinkel et al., 2024) fine-tunes different pre-trained models for synthetic text classification, distilbert-base-cased for the monolingual tasks and distilbert-base-multilingual-cased for the multilingual ones. Moreover, they explore the use of a Random Forest classifier using frozen distilbert-base-cased embeddings concatenated with 20 linguistic and stylistic features. This approach shows how choosing the right PLMs is crucial for better performance in a given task.

Team Mashee_{STA_mono:115} (Rasheed and Zarkoosh, 2024) selects high-quality and low-quality samples using a Chi-square test and adopts the selected samples for few-shot classification using the FlanT5-Large language model. This approach shows how few-shot methodologies can benefit from a careful example selection.

Team TueCICL_{STA_mono:116} (Stuhlinger and Winkler, 2024) uses a Character-level LSTM with pre-trained word2vec embeddings as input to train synthetic text detector. Doing so, they show how one does not necessarily have to use transformers.

Team RUG-1_{STA_mono:117} (Alecakir et al., 2024) combines a linear model with document-level features and token-level features that are first passed through an LSTM. Through this methodology, they leverage both local (token-level) and global (document-level) information to identify human-written and synthetic texts.

Team CUNLP_{STA_mono:unknown}¹ (Aggarwal and Sachdeva, 2024) involved employing a range of machine learning techniques, including logistic regression, transformer models, attention mechanisms, and unsupervised learning methods. Through rigorous experimentation, they identified key features influencing classification accuracy, namely text length, vocabulary richness, and coherence. Notably, the highest classification accuracy was achieved by integrating transformer models with TF-IDF representation and feature engineering. However, it is essential to note that this approach demanded substantial computational resources due to the complexity of transformer models and the incorporation of TF-IDF. Additionally, their investigation encompassed a thorough exploration of various ML algorithms, extensive hyperparameter tuning, and optimization techniques. Furthermore, they conducted detailed exploratory data analysis to gain insights into the structural and lexical characteristics of the text data.

A.2 Multilingual Human vs Machine

Team USTC-BUPT_{STA_Multi:1} (Guo et al., 2024) secured the top position. They initially detect the language of the input text. For English text, they average embeddings from Llama-2-70B, followed by classification through a two-stage CNN. For texts in other languages, the classification problem is transformed into fine-tuning a next-token prediction task using the mT5 model, incorporating special tokens for classification. Their approach integrates both monolingual and multilingual strategies, exploiting large language models for direct embedding extraction and model fine-tuning. This enables the system to adeptly handle text classification across a diverse range of languages, especially those with fewer resources.

Team FI Group_{STA_Multi:2} (Ben-Fares et al., 2024) came in second place. Their methodology began with analyzing latent space distinctions between human and AI-generated texts using Sentence-BERT, hypothesizing that syntax and writing style differences are key. They utilized a hierarchical fusion strategy to adaptively fuse representations from different BERT layers, focusing on syntax over semantics. By classifying each token as Human or AI, their model captures detailed text structures, leveraging the XLM-RoBERTa-Large model for robust multilingual performance.

Team KInIT_{STA_Multi:3} (Spiegel and Macko, 2024) placed third by employing an ensemble of two fine-tuned LLMs (Falcon-7B and Mistral-7B) and three zero-shot statistical methods, using a two-step majority voting system. This unique combination of fine-tuned and statistical methods, complemented by language identification and per-language threshold calibration, showcases their innovative approach to integrating diverse techniques for enhanced classification accuracy.

Team L3i++_{STA_Multi:5} (Tran et al., 2024) explored a comparative study among metric-based models, fine-tuned sequence-labeling language models, and a large-scale LLM, finding LLaMA-2 to outperform others in detecting machine-generated texts. Their methodological diversity and comprehensive analysis underline the strengths of fine-tuning LLMs for complex classification tasks across languages.

Team QUST_{STA_Multi:6} (Xu et al., 2024) employed a fine-tuned XLM-RoBERTa model within a stacking ensemble framework, incorporating the MPU framework and DeBERTa model. Their approach emphasizes the efficacy of model fusion and fine-tuning on a multilingual dataset, highlighting the potential of ensemble strategies in enhancing model performance.

Team AIpom_{STA_Multi:9} (Shirnin et al., 2024) utilized a LoRA-Finetuned LLM for classifying texts as real or fake, achieving notable results with a limited dataset. Their unique approach of using an LLM as a classifier, despite an accidental label swap during training, emphasizes the versatility and potential of LLMs in unconventional scenarios.

Team RFBES_{STA_Multi:10} (Heydari Rad et al., 2024) Both semantic and syntactic considerations were taken into account. For semantic analysis, emphasis was placed on smaller text segments rather than the

¹Team CUNLP submitted results for development set, but no submissions for the test set, resulting unknown valid rank.

entire document, operating under the belief that AI models could produce similarly coherent long texts as humans. To achieve this, the XLM-RoBERTa model was employed. Regarding syntactic analysis, a stacked bidirectional LSTM model was used to categorize texts based on their grammatical patterns using UPOS tags. Interestingly, no significant differences in UPOS tag distribution between AI-generated and human-written texts were revealed by the findings.

Team SINAI_{STA_Multi:21} (Gutiérrez Megías et al., 2024) compared various systems before settling on a fusion model that integrates text with perplexity values for classification. Their comprehensive approach, blending fine-tuning with innovative use of perplexity, offers insightful perspectives on leveraging multiple data dimensions for classification.

Team Unibuc-NLP_{STA_Multi:22} (Marchitan et al., 2024) focused on exploring different methods of layer selection and fine-tuning within a transformer-based architecture. Their pursuit of optimizing layer interactions for classification tasks highlights the importance of fine-tuning strategies in achieving model effectiveness.

Team Werkzeug_{STA_Multi:30} (Wu et al., 2024) applied parametric whitening transformations under a mixture-of-experts architecture to address text embedding anisotropy issues. Their methodological innovation, aimed at capturing a broader range of language styles, underscores the potential of advanced architectures in improving classification accuracy.

Team RUG-5_{STA_Multi:32} (Darwinkel et al., 2024) augmented DistilBERT with an additional layer for classification, exploring linguistic-stylistic features alongside Random Forest classifiers. Their approach of blending traditional ML techniques with PLMs offers a novel perspective on enhancing text classification through feature integration.

Team DUTH_{STA_Multi:33} (Kyriakou et al., 2024) compared machine learning algorithms and LLMs, ultimately selecting a fine-tuned XLM-RoBERTa model. Their comparative analysis provides valuable insights into the effectiveness of different methodologies for text classification tasks.

Team RUG-D_{STA_Multi:39} (Brekhof et al., 2024) used an ensemble of monolingual and multilingual models, testing the performance impact of additional training data. Their ensemble approach and data augmentation strategy highlight the importance of model and data selection in optimizing classification performance.

Team MasonTigers_{STA_Multi:49} (Puspo et al., 2024) experimented with different transformer models and finetuning strategies, showcasing the effectiveness of ensembling and fine-tuning in addressing classification challenges.

Team TrustAI_{STA_Multi:55} (Urlana et al., 2024) focused on fine-tuning the bert-base-multilingual-cased model, demonstrating the potential of pre-trained models in multilingual text classification tasks.

A.3 Multi-way Detection

Team AISPACE_{STB:1} (Gu and Meng, 2024) achieves the highest performance in this subtask by fine-tuning various encoder and encoder-decoder models, including RoBERTa, DeBERTa, XLNet, Longformer, and T5. They augment the data with instances from Subtask A and explore the effects of different loss functions and learning rate values. Based on this analysis, they leverage a weighted Cross-Entropy loss to balance samples in different classes. Furthermore, they use an ensemble of different fine-tuned models to improve the robustness of the system. The weights of the models in the ensemble are assigned based on their performance on each class rather than their performance on the whole accuracy.

Team Unibuc - NLP_{STB:2} (Marchitan et al., 2024) use a Transformer-based model with a peculiar two-layer feed-forward network as a classification head. They also augment the data with instances from Subtask A monolingual dataset.

Team USTC-BUPT_{STB:3} (Guo et al., 2024) first leverage the ‘Llama-2-70B’ model to obtain embeddings of the tokens in the text and then average them across all tokens. Next, they employ a three-stage classification approach using the CNN classifier.

Firstly, they distinguish between human-generated and machine-generated text using the Llama-2-70B model. Secondly, they categorize ChatGPT and Cohere as a single class for a four-class classification, differentiating them from Davinci, Bloomz, and Dolly. Finally, they perform a binary classification